Project 1

Interactive Delhi Metro Route Planner and Visualiser

Design Practices in Computer Science

Project Overview

This project aims to design and implement an **interactive Delhi Metro route planning and visualisation application**. The application should allow users to explore the Delhi Metro network, find routes between stations, and interact with a visual representation of the metro system.
The application should provide an intuitive user interface where users can select a source and destination station and obtain the **fastest available route**. The system should identify and display the stations, metro lines, and interchanges involved in the selected route, along with relevant journey information such as the number of interchanges and estimated travel time.
The project should combine an efficient graph-based route-finding algorithm with an interactive graphical representation of the Delhi Metro network.

1 Core Functional Requirements

1.1 Metro Network Representation

- Representation of the Delhi Metro network as a graph.
- Model metro stations as nodes and connections between adjacent stations as edges.
- Represent different metro lines and interchange stations appropriately.

1.2 Station Search and Selection

- Provide a mechanism to search for metro stations by name or station code.
- Allow users to select a source and destination station.
- Provide appropriate feedback when an invalid or unavailable station or station code is selected.

1.3 Fastest Route Finder

- Find the fastest route between any two selected metro stations.
- Display the complete route from the source station to the destination station.
- Clearly indicate the stations visited along the route.
- Display the metro line(s) used during the journey.

1.4 Interchange Detection

- Identify interchange stations along a route.
- Clearly highlight locations where the user needs to change metro lines.
- Display the sequence of metro lines used during the journey.

1.5 Interactive Metro Map

- Display the Delhi Metro network graphically.
- Show stations and connections between stations.
- Highlight the selected source and destination stations.
- Highlight the currently selected route on the map.
- Visually distinguish interchange stations.

1.6 Map Navigation

- Zoom in and out of the metro map.
- Support panning/navigation across the map.
- Allow users to return to an appropriate default or full-network view.

1.7 Route Reset

- Provide an option to clear the current source, destination, and route.
- Allow users to start a new route search without restarting the application.

1.8 Error Handling

- Provide clear and user-friendly messages for invalid selections.
- Handle cases where no valid route can be found.
- Prevent the application from crashing due to invalid user input.

2 Integration Requirements

- The metro network should be represented using an appropriate **graph data structure**.
- The application should implement an appropriate **shortest-path/route-finding algorithm** for finding the fastest route.
- The implementation should account for **interchange stations and multiple metro lines**.
- Route information obtained from the graph should be integrated with the interactive map so that the selected route is visually highlighted.
- The application should maintain a clear separation between:
  - Metro network data.

  - Route-finding logic.
  - User interface.
  - Map visualization.
- The application should be designed using a **modular architecture**, making it possible to extend the metro network or add new functionality without significantly modifying existing components.
- The system should provide reasonable performance when searching for routes across the complete metro network.

Important

The route planner should not merely display the final route. It should clearly communicate the **route-finding result, including the stations, metro lines, interchanges, and journey information** that explain how the selected route was obtained.

3 User Experience (UX) Requirements

3.1 Simple Route Selection

Provide an intuitive interface for selecting the source and destination stations.

3.2 Interactive Map

Users should be able to visually explore the metro network using zoom and pan operations.

3.3 Clear Route Highlighting

The selected route should be visually distinguishable from the rest of the metro network.

3.4 Station Information

Selecting or hovering over a station should provide useful information such as the station name, metro line, and interchange status.

3.5 Route Information

Route information should be presented in a clear and easy-to-understand format.

3.6 Route Details

The interface should clearly display the selected route, including:

- Source station.
- Destination station.
- Stations along the route.
- Metro line(s).
- Number of interchanges.

- Estimated travel time.

3.7 Responsive Interaction

Map interactions and route selection should be smooth and should not unnecessarily freeze the application.

3.8 Visual Feedback

The application should provide appropriate feedback when a route is being calculated, when no route exists, or when the user changes the selected stations.

3.9 Clean Interface

The application should have a well-organized layout with clearly identifiable controls for station search, route selection, route details, and map navigation.

4 Suggested Additional Features (Optional)

Students may implement additional features beyond the core requirements. Examples include:

- **Alternative Route Generation**
  - Provide multiple valid routes between the selected source and destination.
  - Allow users to switch between available routes.
- **Route Comparison**
  - Allow users to compare alternative routes based on travel time, number of stations, and number of interchanges.
  - Highlight the fastest route, shortest route, or route with the fewest interchanges.
- **Fewest-Interchange Route**
  - Provide an option to find a route that minimizes the number of line changes.
- **Shortest Route**
  - Provide an option to find a route with the minimum number of stations.
- **Favorite Routes**
  - Allow users to save frequently used source-destination pairs.
- **Route Animation**
  - Animate the journey along the selected metro route.
- **Station Details**
  - Display additional information about a selected station, including connected metro lines and interchange status.
- **Dark/Light Mode**

  - Provide different visual themes for the application.

- **Export Route**
  - Allow users to export or save the selected route and its details.

5 Deliverables

- Source code with clear documentation and modular structure.
- Data set from the metro network used by the application.
- Documentation describing the graph representation and route-finding algorithm.
- Documentation explaining the design and architecture of the application.
- Demonstration of the interactive metro map and route-finding functionality.
- All associated assignments and design submissions required for the course.

6 Evaluation Focus

The project should primarily be evaluated based on:

- Correctness of graph representation and route finding
- Accuracy of route and interchange information
- Quality of interactive map visualisation
- Interactivity and responsiveness
- Appropriate use of graph algorithms
- Algorithmic efficiency
- Software architecture and modularity
- UI/UX quality
- Code quality and maintainability
- Regular development and meaningful commits
- Additional functional and algorithmic features